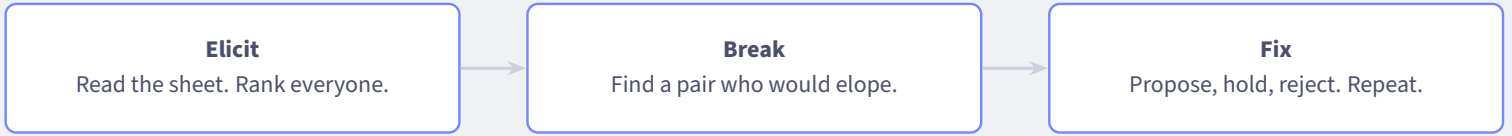
Matching games

Four mathematicians, four physicists, one stable matching



Mathematicians

Gauss				
Noether				
Turing				
Euler				

Physicists

Einstein				
Curie				
Newton				
Feynman				

Most preferred first. These come from the preference sheet, not from me.

While you play

Agree a strict ranking for every mathematician and every physicist. No ties, and everybody ranks everybody on the other side.

Pair everyone up however you like:

	with				with		
	with				with		

Now find two people who would both rather have each other. That pair is a _____, and the matching is _____.

Round 1

Gauss	Einstein
Noether	Curie
Turing	Newton
Euler	Feynman

Round 2

Gauss	Einstein
Noether	Curie
Turing	Newton
Euler	Feynman

Draw each proposal as an arrow. Circle the one held, and cross out the rejections.

Round 3

Gauss	Einstein
Noether	Curie
Turing	Newton
Euler	Feynman

Round 4

Gauss	Einstein
Noether	Curie
Turing	Newton
Euler	Feynman

Stop when nobody is left holding a proposal they have not accepted.

[ ] with [ ]

[ ] with [ ]

[ ] with [ ]

[ ] with [ ]

Check it: take any pair who are *not* matched to each other, and show at least one of them prefers who they have.

Who does better out of this, the proposers or the proposed to? [ ]

What we just did in the room is **Question 1** on the Matching Games page, with three on each side instead of four.

(a)	Define a blocking pair and a stable matching, and write the preferences as maps.	[5]
(b)	Run Gale-Shapley with the mathematicians proposing, showing each step.	[7]
(c)	Write down the resulting matching.	[3]
(d)	Verify it is stable by checking for a blocking pair.	[4]
(e)	Whose optimum is it, and why can the proposing side never do better?	[6]

vknight.org/gt/topics/matching-games.html

Every stage of today, and the part of the notes it was reading.

What we did	Notes	In the textbook
Both sides rank each other	**15.2.1**	Matching Games: Definition: Matching Game
A pair who would rather swap	**15.2.2**	Matching Games: Definition: a blocking pair
Proposing, holding and rejecting	**15.2.4**	Matching Games: Definition: The Gale-Shapley algorithm
Why nobody wants to move	**15.2.3**	Matching Games: Definition: a stable matching
Examinable	**15.2.5, 15.2.6**	Unique matching as output of Gale-Shapley, and reviewer sub optimality, both with proof

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Quick quiz before you go: vknight.org/gt/topics/matching-games.html

Before the next class: read the Matching Games chapter as far as reviewer sub optimality, and attempt Question 1.